Analyzing Voter Difficulty

Datasci 203: Lab 1

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Introduction

Many voters in the U.S. face barriers to voting. To cast a vote, a person may have to surmount obstacles such as long wait times and large crowds at the polls, personal scheduling conflicts, lack of polling places, or difficulty in accessing voting stations, among others. A Pew Research Center survey reveals that while voters consider voting important, only smaller majorities believe it is convenient, straightforward, or exciting. Further complicating the task of voting, many states have passed laws that could make it even harder to vote. Such legislation includes imposing stricter identification requirements and making applying for and receiving a mail-in ballot more difficult. Per Reuters, more than 30 new voting restrictions have been enacted by U.S. states since 2020, and for the 2022 midterms, 51% of eligible voters lived in U.S. states with new restrictions. With the increase of such laws, voting access has become a heated topic of political debate between Republicans and Democrats. Republicans back most of the measures, claiming they are necessary to ensure election integrity, whereas Democrats oppose the measures for making it harder for voters to cast votes.

This analysis explores whether voting difficulty affects political parties disproportionately. Voting difficulty can impact the composition of votes. If one group suffers more from voting difficulty, their interests will not be adequately represented, impacting election results. Significant results from this analysis may indicate a need for reform to support fair and equal representation for all groups. This analysis intends to answer the following research question related to this topic:

> Do Democratic voters or Republican voters experience more difficulty voting?

A study on voting difficulty can motivate political and regulatory reform, leading to more voting power equity. Further insights into voting difficulty may also assist in countering voter suppression tactics. Organizations advocating for voting rights can leverage such knowledge to boost their efforts to make voting more accessible, convenient, and fair.

Data and Methodology

For our analysis, we are using data from the American National Election Studies (ANES) 2022 Pilot Study. The ANES produces high-quality data on voting, public opinion, and political participation to serve the research needs of its community. The 2022 Pilot Study comprised a sample of respondents who are part of YouGov's opt-in online panel, which consists of over a million respondents who complete online surveys for redeemable points. The YouGov panel provides a nonprobability sample, and respondents were chosen from the YouGov panel through sample matching survey participants to U.S. citizens in the 2019 American Community Survey (ACS). The dataset includes 1585 cases.

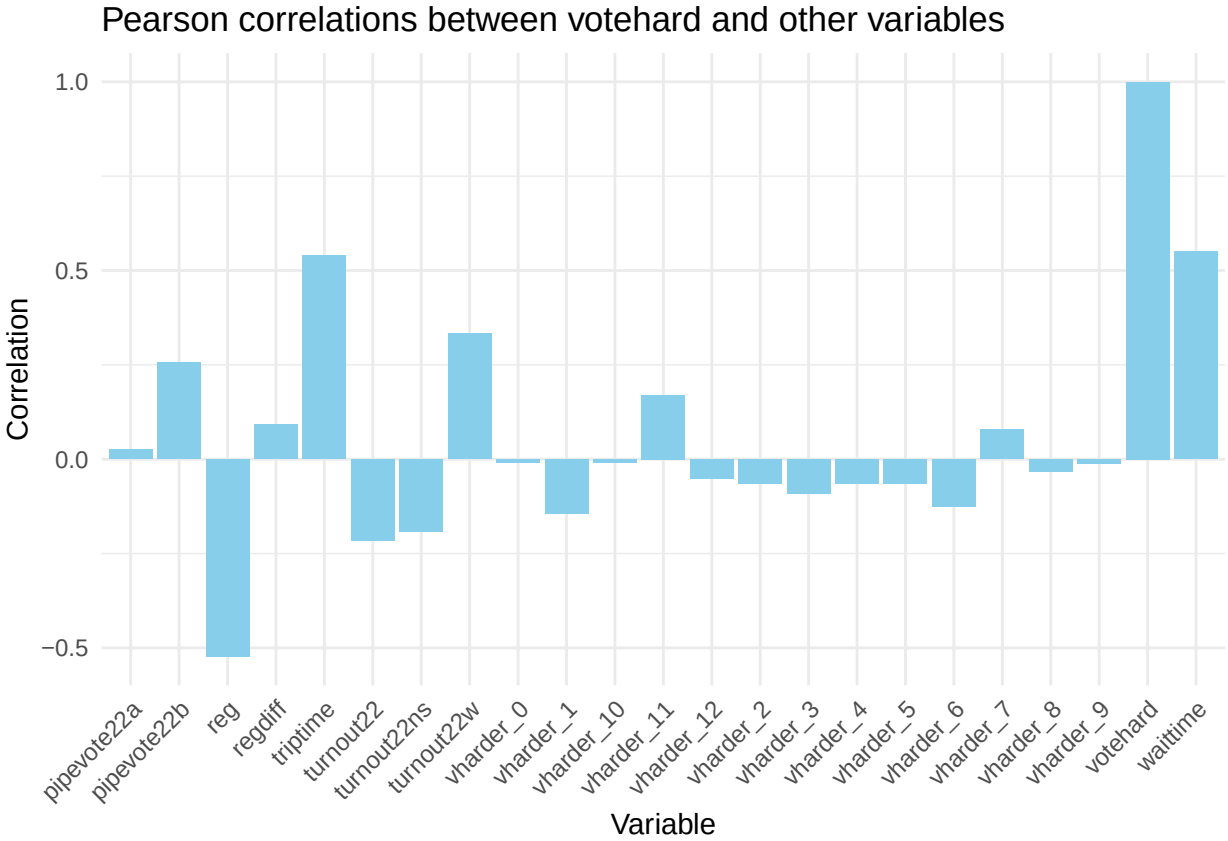
For this project, we conceptualized "difficulty voting" as the anticipated perception of challenges associated with the voting process. Specifically, we used the answers to question 11, "How difficult was it for you to vote?" [votehard], from the ANES 2022 Pilot Study as a measure of voting difficulty. Justification for why we chose this variable can be found later in this section. We define "voter" as those who voted since we are interested in the experience of those who were able to participate in the election. We also define voter this way because of the availability of a voting difficulty measurement for people who voted within the survey. Though it would also be interesting to investigate the voting difficulty of people who were not able to vote, it is out of the scope of this project.

For our analysis, we removed individuals who were not identified as a Republican or Democrat, reducing the number of observations by 201. We identified a person's party affiliation based on their answers to the following questions:

- Generally speaking, do you usually think of yourself as a Democrat, a Republican, an independent, or what?
- Generally speaking, do you usually think of yourself as a Republican, a Democrat, an independent, or what?
- Do you think of yourself as closer to the Republican Party or to the Democratic Party?

Since a person's voting record is also an indication of party affiliation, for those who viewed themselves as independent or something else, we identified them as either a Democrat or a Republican if they voted strictly for one of those parties for at least three offices (i.e., House representative, Senator, governor, or president) in the 2022 midterm election and 2020 presidential election.

We selected 'votehard' from the ANES survey as a comprehensive measure of voting difficulty. This choice is based on its correlation with several objective measures that, according to existing knowledge, are likely to influence the difficulty of voting. For example, 'votehard' positively correlates with both waiting time and trip time, exhibiting a Pearson correlation coefficient greater than 0.5. We chose not to use turnout rate as a measure, since factors beyond voting difficulty could contribute to a low turnout rate.



For the 'votehard' variable, the survey asked participants: How difficult was it for you to vote? Responses are based on five-point Likert scale: 1 equal to "Not difficult at all" and 5 equal to "Extremely difficult". Some participants were not asked this question based on previous responses; respondents had to have answered that they definitely or probably voted in previous questions. Any inapplicable or legitimate skip responses (assigned a value of -1) were removed from the analysis. By limiting our analysis to individuals who believe they most definitely or probably voted in the 2022 midterm election, our analysis includes 523 Republicans and 598 Democrats.

As an alternative target variable, we considered comparing the count of hardships people experienced based on the answers to the following survey question: "Here is a list of things some people say might make it harder for U.S. citizens to vote. Whether or not you were able to vote, please indicate if any of the following made it harder for you to vote in the 2022 general election." However, these variables appeared weakly correlated with the measures noted above.

Hypothesis

Since our target variable is on an ordinal scale and we do not have paired data, there is only one test appropriate for our analysis: Wilcoxon rank sum test (comparisons version).

The null hypothesis of our test is as follows:

> Null Hypothesis: The probability that a Democratic draw ranks higher in voting difficulty than a Republican draw is the same as the probability that a Republican draw ranks higher in voting difficulty than a Democratic draw.

Test Assumptions

In order for a Wilcoxon rank-sum test to produce reliable inference, the following two assumptions must be true:

- measure on (at least) an ordinal scale; and
- IID data—all X_i and Y_i are mutually independent; each X_i is drawn from the same distribution, and each Y_i is drawn from the same distribution.

Our chosen target variable [votehard] is on an ordinal scale; respondents provided answers on a five-point Likert scale.

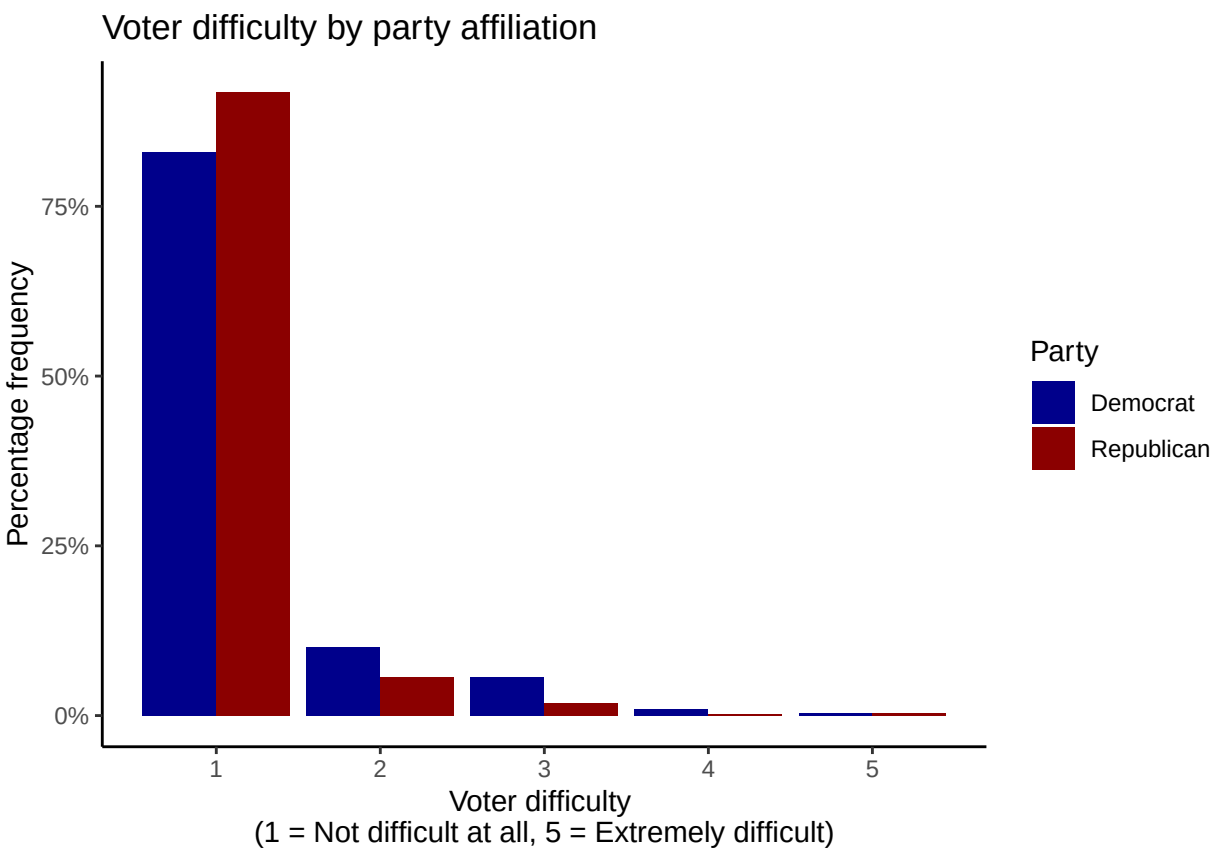
As for IID data, since the YouGov panel comprises participants who have opted in and are completing surveys for rewards, there is a chance for dependency between samples; for example, if two members of one household are on the platform. However, the large, diverse set of YouGov participants and ANES's sample matching method, which aims to represent the population in terms of gender, age, race, and education based on the ACS, may limit the risk of dependence.

Results

```
wilcox.test(data$votehard[(data$party == 1) & (data$votehard != -1)],
            data$votehard[(data$party == 2) & (data$votehard != -1)])
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  data$votehard[(data$party == 1) & (data$votehard != -1)] and data$votehard[(data$party == 2) & (data
## W = 170417, p-value = 8.343e-06
## alternative hypothesis: true location shift is not equal to 0
```

The test yields evidence that the probability that a Democratic draw ranks higher than a Republican draw is different from the probability that a draw ranks lower (p-value $= 8.3430322 \times 10^{-6}$). Said plainly, this study revealed a significant difference in voting difficulty between the two parties, Republican and Democrat. Further, 91.8% of Republicans experienced no difficulty voting, whereas only 82.9% of Democrats experienced no difficulty voting (see below figure). This discrepancy in voting difficulty could potentially lead to bias in election results. We note, however, that the ANES data is not nationally representative, suggesting that our results may not generalize to the U.S. population. We also note that our study is restricted to people who actually voted; there may be difficulties or external factors experienced by nonvoters that are not in the purview of our study.



This study suggests that the voting process may require improvements to ensure all voters can cast their ballots in accordance with their original intent, without being hindered by external factors. These results may be interesting to voting rights advocacy groups wishing to expand access to voting or to policymakers backing political and regulatory reform to curtail imposed voter restrictions.